

For over 300 million years, trees have traded with fungi in underground symbiotic relationships known as mycorrhizas (my-co-rise-ahs). Trees offer some of the carbon they photosynthesize in exchange for nutrients that the fungi extract from the soil. Scientists continue to make new discoveries about this hidden world, including hints that some of the carbon seems to go through the fungal network and into the trees' seedlings.

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A game for 1 to 4 players, ages 10+

In this game, you will play a mature Douglas-fir tree in the Pacific Northwest of North America. Your goal is to establish seedlings and help them successfully grow into trees. At the end of the game, the player who has grown the best set of seedlings with the most valuable symbiotic relationships will win.

Components



4 Quick Sprout guides



4 Player boards



1 Carbon Track board



8 Goal cards

50 wooden black Carbon disks

24 wooden blue Nitrogen cubes

24 wooden orange Phosphorus cubes

24 wooden purple Potassium cubes



24 Seedling tokens (6 per player)



16 Tree tokens (4 per player)



16 Goal Track discs (4 per player)



72 Root tokens (18 per player)



16 Activation tokens (4 per player)



16 Bonus tiles



48 Mushroom tiles

1 Scorepad



1 Solo AI mat



1 First Player marker



1 Carbon Track 2 point token



4 Carbon Track discs (1 per player)



1 Starting Mushroom tile



30 Solo AI cards

Setup

1 **First Player:** Randomly select a first player and give them the First Player marker.

Quick Sprout guides: If this is your first game, hand out the Quick Sprout guides to each player, giving number 1 to the First player, then give the rest out in numerical clockwise order to the rest of the players. (Quick Sprout guides are described on page 4.)

2 **Resource supply:**
Create a shared supply of Carbon, Nitrogen, Phosphourus, and Potassium (C , N , P , and K).



Quick Sprout: Before shuffling tiles for **Step 3**, remove each player's 3 Starting Mushrooms and give them to the players to form each player's starting hand. They are marked 1 through 4 in the top left corner to match their player order. If you have fewer than 4 players, leave any unused player's starting Mushrooms in the stack for shuffling in Step 3.

Example: Player 1's three Starting Mushrooms:



3 **Mushroom supply:** Shuffle the Mushroom tiles and place them in facedown piles where all players can reach them.

Starting hand Mushrooms: Each player draws 3 Mushrooms from the supply, keeping them hidden from other players. *(Skip this if you are using Quick Sprout guides.)*



4 **Forest:** Place the Starting Mushroom tile in the center of the table, leaving room for the players to build the Forest as they add more Mushroom tiles.



5 **Carbon:** Place 1 C from the supply on each of the 4 outer Mushrooms.

Quick Sprout: For **Step 6**, in place of choosing colors, give each player the Player board, Seedling, Root, and Tree tokens in the color that matches the color on their Quick Sprout guides.

6 **Player boards:** Each player takes 1 Player board of a color and puts the following of each in that color on their tree:

- 6 Seedling tokens
- 18 Root tokens
- 4 Tree tokens

7 **Activation tokens:** Each player puts 4 Activation tokens color side up *(ready)* on the top of their board.



8 **Starting resources:** Each player takes the following resources from the supply and places them on the designated spots of their board:

4 Carbon
2 Phosphorus
2 Nitrogen
2 Potassium

9 **Carbon Track:** Place the Carbon Track board off to the side of the play area.

10 **Bonus tiles:** Shuffle the Bonus tiles and deal a number face down to each indicated square on the Carbon Track according to the number of players:

- 2 & 3 players - **3 tiles per square**
- 4 players - **4 tiles per square**

Put the unused tiles back in the box unseen. Put the Carbon Track 2 point token here:



11 **Carbon Track discs:** Place each player's Carbon Track disc on the Start Here space.

12 **Goal cards:** For your first game, or if you are using the **Quick Sprout** guides, use the 3 Goal cards that say, "**Starting Goal**." Otherwise, randomize the Goal cards and deal 3 next to the Carbon Track (randomize which side of each as well).

13 **Goal Track discs:** Place each player's Goal Track disc on the zero of each Goal card.

Quick Sprout guides



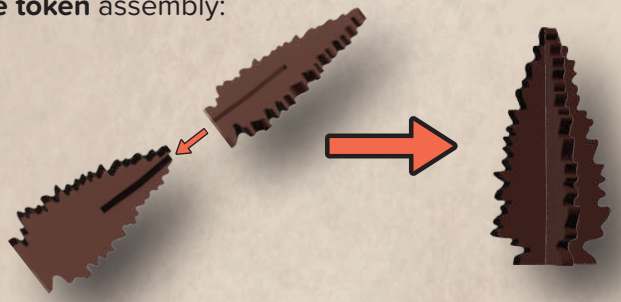
Quick Sprout guides are **not a replacement** for reading the rulebook in order to **understand the rules** and **teach** the game.

They are used to help guide new players through their first 4 turns, and teach core concepts of the game along the way.

Players should take their turns in clockwise order, as they would in a normal game, following the directions on their guide. Everyone should pay attention to the actions other players take during their first 4 turns, as they may show actions and options not given on their own guides.

After everyone has taken the first 4 turns, you may want to review the scoring rules and the Game End trigger and clarify any questions players have. Then continue to play with the players now making their own decisions each turn.

Tree token assembly:



Starting Positions

Quick Sprout: Instead of placing Seedlings and Roots as described below, each player, in clockwise turn order, will follow the instructions for placement on their Quick Sprout guide.

In turn order, players choose a position for their first Seedling in the Forest for free. When it is your turn to choose:

- Place 1 **Seedling** on a corner of the center Earthlover tile.
- Place 1 **Root** on the Earthlover tile, next to that Seedling.



(Player 1 has taken the northeast corner, and Player 2 has chosen the southwest corner.)

- In a **2 or 3 player** game, place Seedlings from the unused player colors in the empty corners of the Earthlover tile (these corners cannot be taken during the game).

Overview

You are playing from the point of view of a mature Douglas-fir tree trying to get your offspring established. You will be placing your Seedling and Root tokens out into a shared Forest area made of Mushroom tiles (**Reproduce** and **Partner**).

Your Roots give you access to trade with Mushrooms for resources (**Activate**).

At the end of the game, your Roots will also score if you have Absorbed enough C **(Absorb)**.

- Seedlings with 1 C score 1 of their Roots.
- Seedlings with 2 C score 2 of their Roots.
- Seedlings with 3 C become a Tree and score all of their Roots (maximum 4 Roots).

When you score a Root, it is worth the points () printed on the Mushroom tile that it's on.

Play continues until one player has advanced their Carbon track disc beyond the last stone on the Carbon track claiming a green reward, which triggers the end of the game. When the end of the game has been triggered, continue the current round until the last player has acted, then starting with the first player, each player takes one final action in turn order.

All points are counted at the end of the game. The player with the most points wins (p. 14).



Each Douglas-fir seed cone represents 1 point.



This Mushroom is worth 3 points.



This Mushroom is worth 2 points.

Turn Summary

1. Perform one Action

- Choose one action to perform from these five:
  - **Activate** (p. 6)
  - **Absorb** (p. 8)
  - **Reproduce** (p. 10)
  - **Partner** (p. 12)
  - **Photosynthesize** (p. 12)

2. Carbon Track (p. 9)

- If you Absorbed 1 or more C this turn, move up the Carbon Track 1 space.

3. Goal Progress (p. 13)

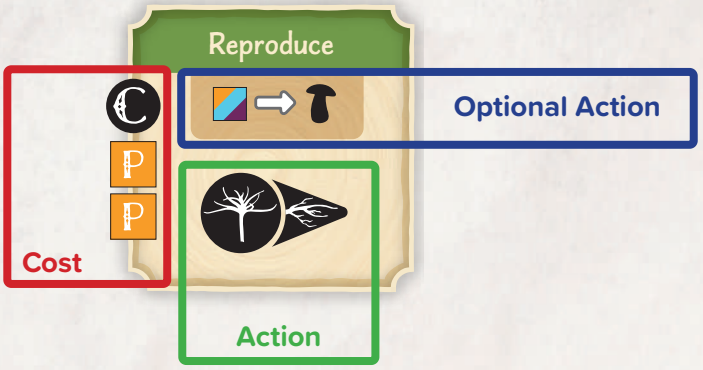
- If you made any progress on a Goal card, mark it with your Goal Track discs on the Goal card.

4. Hand Size

- Draw back up to 3 Mushroom tiles in hand.

Actions

Each action is described in more detail on the following pages. You cannot take more than one action in a turn (unless a Mushroom ability gives you another action). In general, the parts of an action are:



Cost - Required resources, or Activation token flip.
Optional Action - *(brown-shaded boxes)* You ***may*** perform this action in the order shown on your Player board.
Action - The action that will be performed.

Action: Activate

Nitrogen, phosphorus, and potassium are essential to plants. Nitrogen is important for leaves, phosphorus for flowers and seeds, and potassium for transporting sugars.

Throughout the game, you will give C to Mushrooms in exchange for nutrients and other benefits. (Note that some will give you benefits without charging you C and others may require other resources as a cost, spent to the supply.) Mushrooms have a **cost** and an **ability**:



Cost: To Activate a Mushroom, you ***must*** have a Root on that tile. In addition, pay the cost printed on the Mushroom tile:

- : Flip the **ready** Activation token on your Player board that matches the Mushroom type to its **used** side.
- CT : Pay 1 C ***onto the Mushroom tile*** for each CT shown. (*You are trading* C *with the Mushroom for the resources or ability it is giving you.)* A Mushroom tile can have an unlimited amount of C on it.
- : Pay 1 P , N , or K (of your choice) to the supply for each shown.

Ability: Gain the benefit of the ability on the Mushroom tile Mushroom tile abilities are described in detail on p. 24.

Activation tokens

There are 4 Mushroom types in the game. They are grouped based on some basic physical characteristics:



Mushrooms with a ring or cortina on their stem

Mushrooms with pores or teeth

Mushrooms with gills (and no ring)

Mushrooms with other shapes, such as cups and corals

You have 1 Activation token for each type of Mushroom. You can use these tokens to Activate Mushroom tiles and also to move and/or Absorb C (p. 8).

All Activation tokens have a **ready** side (color) and a **used** side (gray):



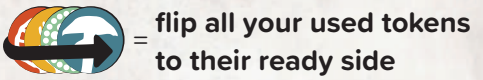
When you use a token, flip it over to its used side. You may not use that token again until you flip it back to the ready side. You will ready all tokens when you Photosynthesize (p. 12). Many Mushroom abilities also ready specific tokens.

Costs usually include flipping an Activation token to its used side with icongraphy like this:



Some abilties allow you to flip an Activation token to its ready side, or all tokens, with iconography like this:





Ongoing abilities

Some Mushrooms have an ability that says, “ONGOING.” You do not need to Activate these ongoing abilities. Instead, they boost another action you might perform. You must have at least 1 Root on them to gain the benefit of their ability, and additional Roots will not increase that benefit.

Activation Example 1: Fly Agaric and 3 other starting Mushrooms have similar effects.



Prerequisite:

- You must have a **Root** on Fly Agaric to Activate it. Brown may Activate the Fly Agaric, but Green cannot.

Cost:

- : Flip your ready to its used side.
- : Place 1 from your Player board onto Fly Agaric.

Ability:

- : Gain 1 and 1 from the supply to your Player board.
- : If your green token is used ( ), flip it to its ready ( ) side.

Activation Example 2: Earthlover is a unique Mushroom that requires no resources to be paid for Activation, but it still uses your action for the turn. It also has an Ongoing ability that replenishes the Forest with when the last is Absorbed.

Cost paid:
No resources.



Ability gained:

and may trade up to any 2

Prerequisite:

- Each player starts the game with a **Root** on Earthlover, so everyone may Activate it.

Cost:

- No resource or activation cost.

Ability:

- : Gain 1 cube of your choice (, , or) from the supply to your Player board.
- : You may trade up to 2 resource cubes from your Player board with any cubes of your choice from the supply. They do not need to match.

Ongoing:

- At the end of each player’s turn, if there is no on any Mushroom tile, place 1 on each of the 4 Mushrooms adjacent to Earthlover (North, East, South, and West).

For all other Mushrooms: See the back of this rulebook for a full glossary of all the Mushroom abilities.

Action: Absorb

Fungi keep most of the carbon that trees give them, but they appear to redistribute some–possibly to keep their trading network strong. This redistributed carbon is small compared to what a tree photosynthesizes for itself, but it may be an important boost for young seedlings.

Seedlings need C for their Roots to score points. Using the Activate action (p. 6), players will often contribute C to Mushroom tiles when Activating them. The Absorb action allows your Seedlings to Absorb this C through one of its Roots.

There may not always be C available on the Mushrooms where you have Roots. You may use the Optional Action to pay extra to move a C before Absorbing it.



Cost:

- : Flip any **ready** Activation token to its **used** side.
- : Pay any 1 cube of your choice to the supply.

Optional Action: Move C

- You can only move the 1 C that you plan to Absorb.
- To move a C, determine how far it must move to get to the Mushroom you want to Absorb it from. C only moves orthogonally from Mushroom to Mushroom (up, down, left, or right–no diagonals).
- For each Mushroom that you move the C to before you Absorb, you must either:
  - : pay any 1 cube of your choice to the supply, or
  - : flip any **ready** Activation token to its **used** side.
- If you move a C more than one space, you may pay any combination of and .
- You do not need to have a Root on the Mushroom where the C starts, only on the Mushroom that you ultimately Absorb it from.
- There are Mushroom powers that let you Absorb 2 C when you Activate them. In that case, you must pay separately for each C that you move before Absorbing it; they do not automatically travel together.

Action: Absorb

- One of your Seedlings Absorbs 1 C from a Mushroom where it has a Root. Take the C off the Mushroom and place it on the Seedling.



  - You cannot pay extra to Absorb more than 1 C on your turn with this Absorb action.
  - Note: The only way to Absorb more than 1 C on your turn is to use a Mushroom power that allows you to do so.
- Move your **Carbon Track** disc up 1 space on the Carbon Track and gain the benefit shown there (p. 9).

***Note:** In our abstracted forest, fungi are each shown on a single tile. In reality, their root systems (known as mycelia (mahy-**see**-lee-uh)) can stretch hundreds or even thousands of meters under the forest floor, connecting distant trees in ways we are unable to depict in this play space. As far as we know, carbon does not pass from one fungus to another, and different fungi generally do not trade with each other.*

Carbon Track

On each turn that you Absorbed C (through the Absorb action or a Mushroom that gives you one or more Absorb actions), move your Carbon Track disc 1 stone up on the Carbon Track.



Immediately take the benefit shown on that stone.

Note: You only **move 1 stone** on the Carbon Track, even if you Absorbed 2 or more C in your turn.

Stones **2**, **4**, and **6** have facedown Bonus tiles on them.

- When you reach a Bonus tile stone on the track, look at the tiles, keep 1, and put the others back facedown.
- Keep your token facedown. You may reveal and then discard it at any time to gain the benefit it shows. If the token gives , reveal it during scoring at the end of the game.
- The types of benefits on the tiles are 2 , 1 , a ready for all your Activation tokens , and each type of resource.

When a player moves past stone **8** (which gives a C) :

- They place their token on 1 of the 5 green spaces and take the reward shown there. Other players who reach the end must choose a different space. The may be added to a player's board as a scoring reminder.
- The game end is triggered. (See Game End, p. 14.)

Example: Brown Absorbs some C this turn, and moves up 1 space. They place 1 Root on 1 of their .



Absorb Example: White would like to Absorb the C from Fly Agaric to their Seedling, but they don't have a Root there. They will use the Optional Action to move it first.



Cost: They flip their ready Activation token, and pay 1 N to the supply from their Player board.

Optional Action: They flip their ready Activation token, and move the C from Fly Agaric to Earthlover.



Action: They Absorb the C from Earthlover to their Seedling.



Carbon Track: They move their Carbon Track disc up 1 stone on the Carbon Track board, secretly look at the 3 Bonus tiles and choose to take the 2 tile, then put the other 2 back.



Strategy Tip: In a sense, the Carbon Track is a race. If you finish first you've likely transfered C more often than the other players. That means more of your Roots are going to score.

Making a Tree

If, after Absorbing, your Seedling now has 3 on it, return the 3 to the supply and replace them with a Tree, which is placed on top of your Seedling token.

Trees give you a scoring boost at the end of the game. A Seedling with 2 scores 2 of its Roots, but a Tree can score all 4 of its Roots (if it has them).



Note that you may make a Seedling into a Tree even if it has only 1 Root, but it is usually efficient to grow Roots around your Trees by the end of the game.

You are limited to making 4 Trees in the game. In the very unusual case that you have made 4 Trees and the game has not ended, your remaining Seedlings may Absorb 3 or even 4 . They will score 1 Root per .

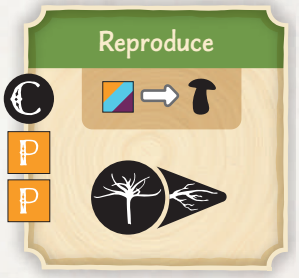
Note: You do not get the Seedling back that you place the Tree on. At maximum, you can only have 6 total Seedlings and/or Trees.



Action: Reproduce

Mature Douglas-firs produce cones that open to release dozens of seeds on the wind. By the end of their first year, successful seedlings usually have a mycorrhizal relationship with at least one fungus. Seedlings that do not have these relationships often fail to thrive.

You start the game with only 1 Seedling in the Forest. This action is how you'll add more Seedlings to the Forest.



Cost:

- Pay to the supply.

Optional Action: Place a Mushroom (p. 11)

- You may choose to pay any 1 to the supply to place 1 Mushroom tile from your hand into the Forest.

Action: Place 1 Seedling with 1 Root

- Place one of your Seedlings from your Player board on **any** unclaimed corner in the Forest.
  - Your seeds are wind-borne: New Seedlings do not need to be near your other pieces.
  - Once placed, Seedlings never move.
  - No other player can use the same corner.
- Add a Root **touching** the Seedling you just placed. The Root must go on one of the Mushrooms that the Seedling is touching.

Placing a Mushroom

The Reproduce or Partner Actions are the only ways to add new Mushroom tiles to the Forest. Both of these actions include the option to pay to place a Mushroom tile. When you do this:

- The Mushroom tile must be placed orthogonally adjacent to at least 1 other Mushroom in the Forest (never diagonally).
- Do not pay the cost shown on the Mushroom. That is the cost to Activate the Mushroom on later turns.
- Remember to always draw your hand back to 3 Mushrooms at the end of your turn.
- When placing your Seedling and/or Root as part of your Reproduce or Partner action, you may place them on the Mushroom you placed this turn, but you are not required to.
- You may only place 1 Mushroom per turn when using the Reproduce or Partner Action. (Note: there are Mushrooms with an ongoing abilities to allow you to place an extra Mushroom.)



Reproduce Example: On their turn, the Green player wants to Reproduce, first placing a Mushroom tile, Peppery Bolete, from their hand next to the Common Deceiver.



Cost: They pay € P P to the supply.

Optional Action: They pay 1 N to the supply and place Peppery Bolete from their hand to the lower right corner of the Forest.



Action: They place a Seedling on an unclaimed corner, and the Root on Peppery Bolete.

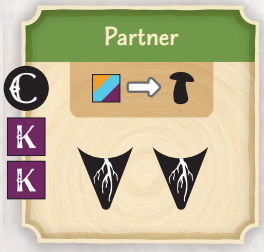


Hand size: At the end of their turn they draw 1 Mushroom from the supply so they have 3 in hand.

Action: Partner

Older trees can have dozens of different fungal species attached to their roots, and each fungus can be attached to many different trees. They form a vast underground network.

Each Seedling comes with just one Root. The Partner action allows you to add Roots to Seedlings and Trees. This gives you access to new Mushroom abilities and their potential points.





Cost:

- Pay to the supply.

Optional Action: Place a Mushroom (p. 11).

- You may choose to pay any 1 to the supply to place 1 Mushroom tile from your hand into the Forest.

Action: 2 Roots

- Place 2 Roots onto Mushrooms.
- Each Root you place **must** be attached to one of your Seedlings or Trees ().
    - These Roots may be attached to the same , but they do not have to be.
- You cannot place a Root on another player's .
- You cannot place a Root on a corner without a Mushroom.
- You can only place 18 Roots at maximum. Most games end long before this.

Action: Photosynthesize

Info box on Photosynthesize. Animals and fungi are all dependent on a food chain that starts with plants photosynthesising carbon dioxide into carbohydrates - the process of converting light energy into chemical energy. Photosynthesizing is the only way to gain in Undergrove.

You will use lots of by activating Mushrooms and making Seedlings and Roots. Taking the Photosynthesize action will replenish your , ready all of your used Activation tokens and discard any unwanted Mushroom tiles from your hand, allowing you to draw fresh ones at the end of your turn.



Cost: No cost.

Action: Gain .

- Take 2 from the supply and add them to your Player board.

Optional Action: + additional for .

- You may pay any number of from your Player board to the supply to gain that many additional .

Action:

- Flip all of your used () Activation tokens to their ready side ().

Optional Action: Discard Mushroom tiles

- You may discard any number of Mushroom tiles from your hand. At the end of your turn, draw Mushroom tiles from the supply until you have 3 in hand.

Goal Cards

Goal cards offer another important source of points. All players may score points for all 3 goal cards. The individual Goal cards and how they score are all described on p. 16.



Goal Tracks: All goal cards have a goal track. Each player starts with a Goal Track disc on each goal at the beginning of the game.

Some goals must be tracked during the game, because they are counting how many times you do a certain thing that isn't otherwise recorded (such as making 6 C in one turn). Others count how many of something (such as Trees) you have at game end. A game-end goal can be checked at any time during the game and marked on its Goal Track.



Some goals may go above 10. In this case, each player has an extra Goal Track disc that they can use to leave on the 10 and then continue counting 11 and beyond:



Score tables: Score tables will show you how many points you score based on your progress on the Goal Track. There are 3 different types of Score tables:

Counter Tables: Award 🌰 dependent on where your Goal Track disc is on the track:



At the end of the game, each player with their disc on the:

- 1 will earn 3 🌰.
- 2 will earn 5 🌰.
- 3 or more will earn 9 🌰.

Competitive Tables: Award 🌰 dependent on how you finished compared to other players on the Goal Track:



In the case of a tie, add the 🌰 for the tied places together and divide them by the number of players rounded down.

- **Example 1:** If 2 players tied for 2nd and 3rd, they'd each score 4 🌰 [(5+3)/2].
- **Example 2:** If 3 players tied for 1st, 2nd, and 3rd, they'd each score 5 🌰 [(9+5+3)/3] rounded down.

Multiplier Tables: Some Score tables just have a multiplier. They will score 🌰 dependent on variables like your progress on the Goal Track:



Example: If your Goal Track disc is at 6 on the Goal Track, you would score 12 🌰.

Game End

Game end trigger: Turns continue until one player triggers the end of the game by reaching the end of the Carbon Track (see p. 9). At that point:

- Finish the round: Continue playing until all players have had the same number of turns.
- Then each player gets one more turn.

Scoring and winning: All points are counted at the end of the game. The player with the most points wins. In the event of a tie, the player with the most Trees wins. If there is still a tie, the player with the most C on their Seedlings is the winner. Score points as follows:

Goal cards: Score as described on each card (p. 16).

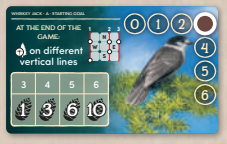
Example: Brown's score for 3 Goal cards:



Brown made 6 C three times in the game for 9 .



Brown had 2 Trees for 2 .



Brown had 6 in 3 different vertical lines for 1 .

	Brown	Tan	Green	Gray
Goal 1	9			
Goal 2	2			
Goal 3	1			
Total Goals	12			

Brown scored a total of 12 for all 3 Goal cards.

Seedlings & Trees : There is space to write the score for each Seedling and Tree on their own line.

- Seedlings with 1 C score 1 Root.
- Seedlings with 2 C score 2 Roots.
- Trees score **all** of their Roots.
- In the **rare** occurrence that all 4 Trees are used Seedlings with 3 or 4 C score 1 Root per C.

The score of a Root is the value shown on the bottom of the Mushroom tile that it is on.

When a Seedling has more Roots than C on it, the player chooses which Roots to score.

Example: Brown's score for their Seedlings & Trees:



1. **Tree** - scores all 4 Roots on 4 Mushrooms for 10 total.

2. **Tree** - scores all 3 Roots on 3 Mushrooms for 12 total, but misses out on scoring Surprise Webcap where it doesn't have a Root. *(Mouse Trich has a variable score dependent on surrounding Gill Mushrooms. It has 3 adjacent Gill Mushrooms, so it scores 2 + 3 .)*

3. **Seedling** - has 2 Roots, but no . It will not score any .

4. **Seedling** - has 3 Roots and 2 , so Brown chooses to score the 2 Roots on Fly Agaric and Peppery Bolete for 6 total *(Peppery Bolete has 3 on it, so it would score 3).*

5. **Seedling** - has 3 Roots, but only 2 , so Brown chooses to score Common Deceiver and Mouse Trich for 8 total.

6. **Seedling** - has 1 Root and 2 , so it only scores Mouse Trich for 5 , but doesn't score Vermilion Waxcap where it doesn't have a Root.

	Brown	Tan	Green	Gray
1	10			
2	12			
3	0			
4	6			
5	8			
6	5			
Total	41			

Bonus tiles & resources: Write down the score from any collected on Bonus tiles. You will also gain 1 point for every 2 unused resources (, , , or) left on your Player board at the end of the game.

If you claimed the token from the end of the Carbon Track, count those points here.



Example: Let's finish Brown's scoring by writing down their points from Bonus tiles and leftover resources:



- 2 Bonus tiles with 3 for **3 points**.
- 7 resources for **3 points**.

	Brown	Tan	Green	Gray
Goal 1	9			
Goal 2	2			
Goal 3	1			
Total Goals	12			
	10			
	12			
	0			
	6			
	8			
	5			
Total	41			
Bonus Tiles	3			
	3			
TOTAL	59			

Finally, add the scores for Total Goals, Total , Bonus tiles, and resources. Brown scored a Total of 59 for this game.

Goal Cards: Glossary

Silver-haired Bat - A: Track how many times you've made 6 or more C in a turn. To do this, you'll generally have to pay 4 or more N to the supply when you take the Photosynthesize action. You can also count C from bonus tiles or ongoing Mushroom abilities.

Silver-haired Bat - B: Each time you Photosynthesize and pay N to the supply to gain additional C, move your disc up the Goal Track 1 space for each N you paid.

Whiskey Jack - A: This goal encourages you to spread out your Seedlings & Trees ( ) on vertical lines from West to East (N, S, E, and W are printed on the starting Mushrooms). You don't want your pieces to be above or below each other on the same vertical line; you want to be on many different lines.



Example: This player has 6 in 5 different vertical lines, so they score 6 .

Whiskey Jack - B: Count how many unique Mushrooms your are touching. Each Mushroom counts only once, no matter how many of your are touching it.

You do not have to have a Root on a Mushroom for it to count.



Example: This player has 6 touching 7 unique Mushrooms.

Red Vole - A: Simply count how many Trees you've grown.

Red Vole - B: Count how many Seedlings & Trees you have with 2 or more Roots.

Douglas-fir Spittlebug - A: This goal tracks how many times you place 2 Roots on the same type of Mushroom in a single turn. The Mushroom's type is indicated by the symbol to the left of its name (/ / /). The 2 Roots may be on the same Mushroom or 2 different Mushrooms of the same type.

Douglas-fir Spittlebug - B: Count how many Roots you've placed.

Flying Squirrel - A and B: These goals look at the point value of the Mushroom that each Root is on and whether that value is EVEN or ODD.

A player may count a Mushroom with multiple Roots more than once; **the count is by Root**.

Some Mushrooms have variable points, and their point value will change throughout the game. Only their value at the end of the game will count for these goals.

Red Crossbill - A and B: These goals look at the name of the Mushroom that each Root is on. Mushrooms are also marked with for names with a color, and for names with a person or animal if there is ever any question.

A player may count a Mushroom with multiple Roots more than once; **the count is by Root**.

Banana Slug - A and B: These goals look at each Seedling and each Tree and want a specific number of Mushroom types touching the piece (/ / /). The 4 types of Mushroom are also shown on page 6.

You do not have to have a Root on the Mushroom for it to count.

Douglas Squirrel - A and B: These goals look at each Seedling and each Tree and want at least 2 Mushrooms with a certain ability touching the piece.

Mushrooms that Ready an Activation token include at least one of these symbols: .

You do not have to have a Root on the Mushroom for it to count.

Whispering Woods: Goal Cards

This expansion can be found at: **alderacstore.com**

Douglas-fir Moth - A: Each Root can only count towards 1 set.

Note: Some Mushrooms have variable points, and their point value will change throughout the game. Only their value at the end of the game will count for this goal.

Douglas-fir Moth - B: This goal counts how many Seedlings & Trees you have on these 12 corners (red dots), one ring out from Earthlover (the center Mushroom):



Sooty Grouse - A: Move your disc up the Goal Track any time you take the Absorb action or Activate a Mushroom that allows you to Absorb and move 1 two or more spaces. If you move a second 2 or more spaces in the same turn, you will move your disc up the track an additional space.

Sooty Grouse - B: This goal counts any time that you ready all 4 Activation tokens in a turn by any method—the Photosynthesize action, activating Mushrooms, or by using a Bonus tile.

Only used tokens can be readied, so you must have 4 used Activation tokens and ready them all in a single turn to count.

Solo Rules

In Undergrove Solo, you will play against 2 other trees trying to establish their Seedlings in the Forest. They will share a single AI deck that controls their actions.

Solo Setup

- **Main setup:** Referring to the setup instructions on p. 2-3, set up a 3-player game between you (Green) and the AI (Brown and Tan), with the following exceptions:
  - **Steps 6-9:** The AI will play as Brown and Tan, but only needs Seedlings. It will not use Roots, Trees, Activation Tokens, or a Player board. It does not get starting resources or Mushroom tiles.
  - **Step 12:** Set this up as normal. The AI does use their Brown and Tan Carbon Track discs.
  - **Steps 13-14:** Randomly choose 3 Goal cards. The AI will not compete on these goals. If you draw any goals that compare players to each other (with a score table that says 1st/2nd/etc.), replace them with other goals.
  - **Starting Position:** You claim a position as normal. The AI does not claim starting positions. Leave the unclaimed corners of the Earthlover open–do not block them with the unused player color.
- **Additional solo setup:**
  - **AI Mat:** Use the provided mat to organize the AI materials.
  - **AI Decks:** Shuffle the AI cards in 2 separate decks (A & B). Stack them in one facedown pile on the AI mat, with deck B on the bottom and deck A on the top.
  - **AI focus elements:** Pick up 1 of each element (N , P , and K) and randomly select one. Place it on the AI mat as Brown's focus element. Repeat for Tan, again randomly selecting 1 of the 3 elements. (Both AI players might end up with the same focus element.)

Solo Gameplay

The AI is the first player, and then will alternate its turns with your turns.

On the AI's turn, they draw the top card from their deck and execute each action on it (*explained in the next section*).

On your turn, play as normal.

- If you Activate a Mushroom that gives other players resources, give both Brown and Tan resources as if they were players, adding them to their resource areas of the AI mat.
- If an AI color would ever gain a cube of its choice, it always chooses its focus element.
- If you are the first to take a Bonus tile from a space on the Carbon Track, shuffle the remaining tiles before placing them facedown on the track.

Either you or one of the AI colors may trigger the **game end** by reaching the end of the Carbon Track. Follow the game end rules as if the AI was the first player:

- **If you trigger the end**, the AI takes one more turn, then you get one more turn. If the AI reaches the end of the track after the game end has already been triggered, it will take, in order:



1st - 1 € and 2 of their focus element.



2nd - Claiming the token



3rd - 3 €.

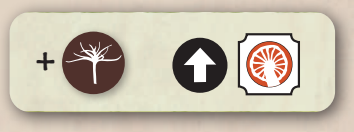
- **If the AI triggers the end**, they always take their first choice spot shown above. You get one more turn in that round, then you and the AI both get one additional turn.

AI Cards

Each AI card has several actions on it. Perform each one in order from top to bottom.



Each action usually has two parts. The first part tells you **what** action the AI is going to take and the second part will tell you **where** that action will take place in the Forest (All the AI Actions are covered starting on the next page).



This action is telling you place a Brown Seedling onto the top-most Mushroom in the Forest.



This action is telling you place 2 € from the supply onto the left-most Mushroom in the Forest.

Unable to fulfill action: Whenever the AI cannot fulfill one or more of the actions on their card, you take a N, P, or K of your choice. You gain only one resource on the AI's turn, no matter how many actions they could not carry out.

Location

The arrows, followed by a Mushroom type icon (/ / /), represent a location in the Forest:



Top-most Bottom-most Left-most Right-most





Example: The Top-most Mushroom in the Forest is the Common Deceiver.

Location Ties: When 2 or more locations are tied, break ties in reading order (left to right / top to bottom). For example if 2 Mushrooms are both Top-most, then the Mushroom on the left wins the tie–or if two Mushrooms are both Left-most, then the Mushroom on top wins the tie.





Example: There are 2 Top-most Mushrooms. The tie is broken left to right, so the action is taken on Painted Suillus.

Location cannot be used: If the first Mushroom picked by the Location info cannot be used, find the next available Mushroom of the same type as if that unusable Mushroom did not exist. If there is no usable location, the AI skips the action. See **Unable to fulfill action** on p. 18.

AI Actions



Place a Mushroom tile: The AI places the top Mushroom tile from the deck if there is a space available.

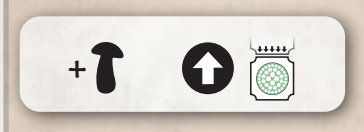


The location iconography will have additional information showing which edge of the Mushroom to place the new Mushroom tile on:



This action is telling you to place a Mushroom tile on the top-most type Mushroom, along its bottom edge.

If the first Mushroom location indicated does not have that edge available, move along to the next Mushroom of that type to see if it has the indicated edge available. If its edge is not open, continue on in reading order to see if any of the Mushrooms of the indicated type have that edge available.





Example: The Peppery Bolete is the top-most Mushroom, but its top edge is blocked. The new Mushroom is placed on the top edge of the Suede Bolete instead.

If no Mushroom of the type you're looking for has the correct edge available, the AI does not place a Mushroom. See **Unable to fulfill action** on p. 18.



Place a Seedling: The AI places a Seedling of the indicated color on a Mushroom if there is a space available.

Find the first indicated Mushroom that has a corner space available. On that Mushroom, find the corner with the most connected Mushrooms. Place a Seedling of the color shown on the AI card on that corner. If two corners are tied, choose by reading order: left to right, then top to bottom.



Example: The Peppery Bolete is the top-most Mushroom, but its corners are full. The new Tan Seedling will go on the Suede Bolete instead, taking the corner it shares with Scaly Vase.

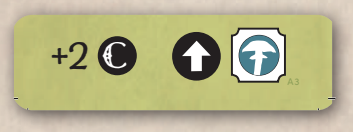


If there is no matching Mushroom with a corner available, the AI does not place a Seedling. See **Unable to fulfill action** on page 18.



Add Carbon : The AI puts a on a Mushroom in the Forest from the **supply** (never its own board).

Use the location info to find the **first** Mushroom of the indicated type. Add the indicated number of to that Mushroom.

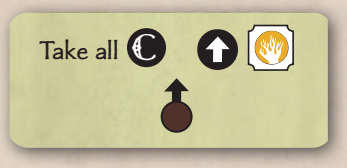


This action will add 2 to the Top-most type Mushroom.



Take Carbon : The indicated AI color takes from a Mushroom and adds it to the resource area of their board. They then move up the Carbon Track 1 step.

Use the location info to find the **first** Mushroom of the indicated type that ***has*** ***on it.*** The AI takes the number of shown on the AI card and adds the to its board. This is an important source of points for the AI—do not return the to the supply!



In this Action the Brown AI player (color is indicated in the circle with the up arrow) takes all the from the top-most Mushroom that has any on it and place it on the Brown board. The Brown player moves up the Carbon Track.

If there are no matching Mushrooms with on them, the AI does not gain any. See **Unable to fulfill action** on p. 18.

However, the indicated AI color still moves up the Carbon Track even if there was no to take.

As the AI players move up the Carbon Track, they take the rewards shown on the spaces, except for free Roots (you gain free Roots from the Carbon Track as normal). Add them to the Player board of the AI player that took the .

When the AI gains a Bonus tile, it takes the top tile in that stack. , , , , and all count at the end of the game. The bonus tiles that Ready Activation tokens have no effect.

When the AI gains a Bonus tile you may choose to reveal it to see what benefit the AI gained, or you may choose to leave it hidden and only discover its benefit at the end of the game.

Solo Scoring

You score as normal, as described on pages 14-15.

The AI scores by player color, getting one score for Brown and one score for Tan, as follows:

AI Scoring:

- **Focus element:** Each AI player gains 1 point for each resource they have gathered that matches their focus element. Record this in the goals section of the scorepad.
- **Seedlings:** By player color, count each of the AI's Seedlings. Score each AI Seedling as you would normally score a Tree with 4 Roots.
- **Difficutly Mode:** Pick a difficulty level by crossing out some, or none, of the AI Seedling scores:
  - **Easy Mode:** Each AI color keeps their 4 best Seedlings. Cross out the rest.
  - **Medium Mode:** Each AI color keeps their 5 best Seedlings. Cross out the rest.
  - **Hard Mode:** Both AI colors score all of their Seedlings.
- **Bonus tiles, Carbon Track, and resources:** The AI colors score like a human player. Count up the Bonus tile points and leftover resources in each resource area of the AI mat. Add the 2 points on the Carbon Track if one of the AI tokens reached the end. Note that this is in addition to the focus element points above (i.e. each AI color scores a total of 1.5 points per focus element it took).

You win if you have a higher score than both of the AI colors.

Added Difficulty

The AI decks intentionally have 3 **The AI takes no actions this turn** cards in them. You may choose to remove 1 or more of them from the AI decks. The more you remove the harder the game becomes.

Strategy Tips

Good for 1-4 players

Placing Roots: Placing Roots - If you have Roots on too many Mushrooms of one type, you won't be able to use them all, because your Activation token will be exhausted on the first one used. Diversify!

Absoring carbon: Try to make a good match between your Roots and your C absorption by the end of the game–you don't want a lot of Roots on Seedlings that don't have C or C on Seedlings that don't have Roots. If you have to pick and choose, Absorb C where you have the most valuable Roots.

Watch other players: Be aware of where other players are with their resources and activation tokens. Although the timing will not always work out, putting C on a mushroom that you might want to absorb, when another player who has nearby access to it, is better done when they have used their activation tokens, or have no resources to pay the cost of absorbing.

High point Mushrooms: When placing a mushroom that has high point value you may want to consider whether you can get an additional seedling on it. Consider where other players are in terms of the resources they need to drop seedlings on your mushrooms.

Bonus tiles: Your bonus tiles can often have good surprise value. If another player has placed C or a mushroom thinking you cannot currently access it, then flipping over one or two bonus tokens and swooping in can be profitable.

Carbon Track: Good position on the carbon track is important not just because it means you have transferred more C but because it gives you control over the bonus tokens and the end of game trigger.

Designer Notes

After working together on Mariposas, we knew we wanted to work together again. When Elizabeth mentioned that she'd been wanting to do a game about mushrooms–and specifically the fact that trees trade resources with mushrooms underground–it felt worth pursuing.

Mycorrhizal symbiosis is truly one of the wonders of the natural world. Plus, as a trading system, it automatically suggests a mechanic for a board game. But beyond all of that, it's about connections. In a world that feels increasingly fragmented, the fact that there are unseen partnerships under our feet is a comfort, a message that maybe everything really is more connected than it seems.

From both of us: It has been an absolute joy to work with the whole crew at AEG. So much effort goes into making a game actually come to life, and you all do it with care and enthusiasm both for the product and for all the people behind it.

Thanks to the many, many playtesters on the next page, who helped us run through hundreds of games of Undergrove before it was done. We literally couldn't have made this game without you.

And thanks to the scientists who have done the difficult work of bringing these hidden systems to light, and those who continue to do so. Particular thanks to Monika Gorzelak, who spent time with us early in our design process as we were trying to decide on the underlying structure of this game.

From Mark: I'd like to thank everyone that has worked on this game. It has been a joy to work with Elizabeth, and many thanks to John Zinser for suggesting it. And of course my fantastic Vancouver Island people - and especially my wife Liz, together with Chris, Damon, Tim, Mike and Kyle for their support over the many years we have play-tested.

Finally I'd like to remember my Highlander Games Dundee crew - Gary C, Silas, Clark, GT, Craig, Specs, Gary H, Paul, Andy, Joules, Dougie, and Graeme. It is true to say they rekindled my love of board games, and for the first few years between ending my time as a ranger and switching to a new career, they were a fantastic support.

From Elizabeth: Thanks to the many designers in the Washington, DC area who create such a community ethos of wanting to see everybody succeed. And especially to Matthew, Dominic and Ben, my weekly kitchen table crew who keep me moving forward.

Thanks to Matt, my partner and hiking buddy, who got me hooked on fungi in the first place two decades ago. He picks up an awful lot of slack when games take over my entire schedule–and brings me mushrooms when I don't get to go on the hunt myself.

A Note on Vocabulary:

Technically, a mushroom is only the aboveground, spore-producing part of a much larger fungal organism. By this definition, trees don't care about mushrooms at all—they care about the underground part of the fungus that is there year-round, exploring for resources and trading with them. However, we humans do love mushrooms...so we've taken the liberty of focusing the tiles on the mushrooms that each fungus produces, even if it's for just a few weeks out of the year. We trust that the rest of the game will remind you that there is much more to these fungi than what you see aboveground.



Resources

- To find a local mushroom club in North America: NAMyco.org/clubs
- To learn more about the importance of mycorrhizal networks: www.spun.earth
- To see some of the work on Douglas-fir ecosystems that helped to inspire this game: MotherTreeProject.org

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Convention Mushroom Team: Teela Dancey, Justin McQuaig, Adelheid Zimmerman, Ryan & Delaina Dancey



Mushroom Iconography

Remember that you must have a Root on a Mushroom to Activate it!

Earthlover
More detail on p. 7

: You may trade up to 2 resource cubes with any cubes of your choice from the supply.

Costs
Left side of the ability box

Flip your matching token to used.

Place 1 on the Mushroom.

Pay 1 , , or to supply.

Resources
Gain the resources shown

Nitrogen.

Gain 1 , , or

Phosphorus.

Potassium.

All OTHER players gain the resource shown.

Ready
Flip your token to its ready side

Take Another Action
Take any of the 5 Actions from your player board, paying all costs.

Absorb
Follow the rules for Absorb action

- must be Absorbed through a Root.
- You may pay extra or to move to that Root.
- Move up 1 space on the Carbon Track.

If you Absorb 2 :

- They may go to different Seedlings.
- You still move up just 1 space on the Carbon Track (not 2).

Absorb " from HERE": The must start on this Mushroom. (This may include you paid to Activate the Mushroom.) You may pay to move it to a different Mushroom and Absorb it there.

Absorb "ANY ": Each may start on any Mushroom. If Absorbing 2 , they may start on 2 different Mushrooms.

Ongoing
These powers boost one of your 5 actions for the rest of the game, starting on the turn after you place a Root on them.

Having a second Root does not double the power.

+1 On the action noted (**Reproduce** or **Partner**), place 1 additional Root. On Reproduce, the Root must be attached to the Seedling you placed.

+1 The Photosynthesize action now gives you 3 instead of 2 .

2nd On the action noted (**Reproduce** or **Partner**), if you pay for the option to place 1 Mushroom tile, you may place a 2nd Mushroom tile for free.

Copy
Select any Mushroom of the matching type and carry out its action, as if it were printed on the copycat Mushroom. This includes:

- Pay the cost of the copied Mushroom **on the copycat**.
- Pay any resource cost to the supply.
- Do not flip another activation token.
- If copying an "Absorb from HERE" action, Absorb from the copycat.

Variable Scoring

At game end, score points only for spaces with a cube left on them. Spaces must be filled from left to right.

Yellow Veiled Amanita: Once full, the extra resource remains in the supply.

2 points, plus 1 point for each adjacent Mushroom of the indicated type. Diagonals do not count.

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